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On the tails of Dickman-like perpetuities

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based on joint work with Oleh Iksanov (Kyiv, Ukraine)

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Let $(M_k, Q_k)_{k \geq 1}$ be independent copies of an $\mathbb{R}^2$ -valued random vector (M, Q) . Whenever the random series

$$Q_1 + \sum_{k \geq 1} M_1 \cdot \dots \cdot M_k Q_{k+1}$$

converges a.s., its sum that we denote by Z is called *perpetuity*.



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converges a.s., its sum that we denote by Z is called *perpetuity*.

A theory of perpetuities accumulated up to 2016 –
[Buraczewski, Damek & Mikosch](#) (2016); [I.](#) (2016).

An incomplete list of more recent contributions

- Tail behavior in a heavy tailed scenario:
[Buraczewski & Damek](#) (2017); [Kevei](#) (2017);
[Damek & Kolodziejek](#) (2020)
- Recurrence vs. transience of $X_{n+1} = M_n X_n + Q_n$:
[Alsmeyer, Buraczewski & I.](#) (2017); [Alsmeyer & I.](#) (2023)
- Limit theorems for $Q_1 + \sum_{k \geq 1} (M_1 \cdot \dots \cdot M_k)^b Q_{k+1}$ and
 $Q_1 + \sum_{k \geq 1} e^{-bk} M_1 \cdot \dots \cdot M_k Q_{k+1}$ as $b \rightarrow 0+$:
[I., Nikitin & Samoilenko](#) (2021); [I., Marynych & Nikitin](#) (2023)



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Goldie & Grübel (1996) proved that if

- $|Z| < \infty$ a.s.;
- M and Q are independent;
- $\mathbb{P}\{M \in (0, 1]\} = 1$;
- the function $\delta \mapsto \delta^{-1} \mathbb{P}\{M \in [1 - \delta, 1]\}$ is bounded away from 0 and ∞ for small positive δ
(if $\mathbb{P}\{M \in dx\} = \text{const } x^{\alpha-1}(1-x)^{\beta-1} \mathbb{1}_{(0,1)}(x)dx$, then **this holds** iff $\alpha > 0$ and $\beta = 1$!);
- the constant $b := \sup\{x : \mathbb{P}\{Q \geq x\} > 0\}$ satisfies $b \in (0, \infty)$,

then

$$-\log \mathbb{P}\{Z > t\} \sim b^{-1} t \log t, \quad t \rightarrow \infty,$$

that is, the right distribution tail of Z exhibits a **Poisson-like decay**.



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What about **precise**, rather than logarithmic, asymptotic under comparable assumptions?



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Standing assumptions:

- M and Q are independent;
- M has the beta distribution with parameters $\alpha > 0$ and 1, that is, $\mathbb{P}\{M \in dx\} = \alpha x^{\alpha-1} \mathbb{1}_{(0,1)}(x) dx$;
- $\mathbb{E}[\log^+ |Q|] < \infty$ and
- $b = \sup\{x : \mathbb{P}\{Q \geq x\} > 0\} \in (0, \infty)$.

We call the corresponding perpetuity Z_α a **Dickman-like** perpetuity.
Why?



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If $Q = 1$, the variable $Z_1 - 1$ has the **Dickman** distribution.

If $Q = 1$ and $\alpha > 0$, the distribution of $Z_\alpha - 1$ is sometimes called **generalized Dickman distribution**.



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The distribution of a real-valued random variable X is called **selfdecomposable** if, for each $\delta > 0$, $z \mapsto \mathbb{E}[e^{izX}]/\mathbb{E}[e^{i\delta zX}]$, $z \in \mathbb{R}$ is the characteristic function of a proper probability distribution.

Each nondegenerate selfdecomposable distribution on $\mathbb{R}$ is absolutely continuous w.r.t. Lebesgue measure and unimodal ([Yamazato \(1978\)](#)).



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Each nondegenerate selfdecomposable distribution on $\mathbb{R}$ is absolutely continuous w.r.t. Lebesgue measure and unimodal ([Yamazato \(1978\)](#)).

Proposition. Let $\alpha > 0$ and Z_α be the Dickman-like perpetuity, that is, it corresponds to (M_α, Q) satisfying the standing assumptions. Then

(a) $Z_\alpha - Q_1 = \int_{[0,\infty)} e^{-y} dA(y)$, where A is a compound Poisson process with the jumps $Q_1, Q_2, \dots$ and the jump times $-\log M_{\alpha,1}, -\log M_{\alpha,1}M_{\alpha,2}, \dots$; in particular, the distribution of $Z_\alpha - Q_1$ is selfdecomposable ([Jurek & Vervaat \(1983\)](#));

(b) $\mathbb{E}[e^{sZ_\alpha}] < \infty$ for all $s \geq 0$ and

$$\mathbb{E}[e^{sZ_\alpha}] = \mathbb{E}[e^{sQ}] \exp \left(\alpha \int_0^s \frac{\mathbb{E}[e^{yQ}] - 1}{y} dy \right), \quad s \geq 0.$$



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Each nondegenerate selfdecomposable distribution on $\mathbb{R}$ is absolutely continuous w.r.t. Lebesgue measure and unimodal ([Yamazato \(1978\)](#)).

Proposition. Let $\alpha > 0$ and Z_α be the Dickman-like perpetuity. Then

(c) For $t > 0$,

$$\alpha^{-1}tq_\alpha(t) = \mathbb{P}\{Z_\alpha > t\} - \mathbb{P}\{Z_\alpha - Q_1 > t\},$$

where q_α denotes a continuous on $(0, \infty)$ density of the distribution of $Z_\alpha - Q_1$.

Furthermore,

$$\alpha^{-1}tq_\alpha(t) \sim \mathbb{P}\{Z_\alpha > t\}, \quad t \rightarrow \infty.$$



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For large t , the equation

$$\alpha \frac{\mathbb{E}[e^{sQ}] - 1}{s} = t$$

has a unique positive solution to be denoted by $s_\alpha(t)$.

Put

$$p_\alpha(t) := \int_{(-\infty, b]} q_\alpha(t-y) d\mathbb{P}\{Q \leq y\}, \quad t \in \mathbb{R}.$$

Then p_α is a continuous on (b, ∞) density of the distribution of Z_α .



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Put

$$p_\alpha(t) := \int_{(-\infty, b]} q_\alpha(t - y) d\mathbb{P}\{Q \leq y\}, \quad t \in \mathbb{R}.$$

Then p_α is a continuous on (b, ∞) density of the distribution of Z_α .

Theorem. Under the standing assumptions, as $t \rightarrow \infty$,

$$p_\alpha(t) \sim \frac{1}{\alpha(2\pi b)^{1/2}} t^{1/2} s_\alpha(t) \exp\left(-ts_\alpha(t) + \alpha \int_0^{s_\alpha(t)} \frac{\mathbb{E}[e^{yQ}] - 1}{y} dy\right)$$

and

$$\mathbb{P}\{Z_\alpha > t\} \sim \frac{1}{\alpha(2\pi b)^{1/2}} t^{1/2} \exp\left(-ts_\alpha(t) + \alpha \int_0^{s_\alpha(t)} \frac{\mathbb{E}[e^{yQ}] - 1}{y} dy\right).$$



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Let X be a real-valued random variable with $\mathbb{P}\{X > 0\} > 0$ and $\Psi(s) := \log \mathbb{E}[e^{sX}] < \infty$ for all $s \geq 0$. Assume that the distribution of X admits a density p . For large t , let $s(t)$ denote a unique positive solution to $\Psi'(s) = t$. Then

$$p(t) = \exp(-ts(t) + \Psi(s(t)))J(t)$$

for appropriate function J and large t .

Put $I(t) := \sup_{s \geq 0} (st - \Psi(s))$ for $t > 0$. The function I is the **Legendre transform** of the distribution of X . Since $I(t) = s(t) - \Psi(s(t))$ for large t ,

$$p(t) = \exp(-I(t))J(t)$$

for large t .



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$$p(t) = \exp(-ts(t) + \Psi(s(t)))J(t)$$

$$p_{\alpha}(t) \sim \frac{1}{\alpha} ts_{\alpha}(t) \exp\left(-ts_{\alpha}(t) + \alpha \int_0^{s_{\alpha}(t)} \frac{\mathbb{E}[e^{yQ}] - 1}{y} dy\right) \\ \times \frac{1}{(2\pi bt)^{1/2}}, \quad t \rightarrow \infty$$

The main contribution of the theorem is a **universal asymptotic behavior of the second factor**

$$J(t) \sim \frac{1}{(2\pi bt)^{1/2}}, \quad t \rightarrow \infty.$$

There is nothing special in the factor

$$\exp(-ts(t) + \Psi(s(t))) \\ \sim \frac{1}{\alpha} ts_{\alpha}(t) \exp\left(-ts_{\alpha}(t) + \alpha \int_0^{s_{\alpha}(t)} \frac{\mathbb{E}[e^{yQ}] - 1}{y} dy\right), \quad t \rightarrow \infty.$$



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EXAMPLE 1. Under the standing assumptions, as $t \rightarrow \infty$,

$$s_\alpha(t) \sim b^{-1} \log t \quad \text{and} \quad ts_\alpha(t) - \alpha \int_0^{s_\alpha(t)} \frac{\mathbb{E}[e^{yQ}] - 1}{y} dy \sim b^{-1} t \log t.$$



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EXAMPLE 1. Under the standing assumptions, as $t \rightarrow \infty$,

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EXAMPLE 2. In addition to the standing assumptions, assume that either $\mathbb{P}\{b - Q \leq x\} \sim x^\theta \ell(x)$ as $x \rightarrow 0+$ for some $\theta \geq 0$ and some ℓ slowly varying at 0 or $\mathbb{P}\{Q = b\} > 0$ (in which case we put $\theta = 0$ in formula below). Then, as $t \rightarrow \infty$,

$$\begin{aligned} s_\alpha(t) &= b^{-1}(\log t + (1 + \theta) \log \log t) + o(\log \log t) \quad \text{and} \\ ts_\alpha(t) - \alpha \int_0^{s_\alpha(t)} \frac{\mathbb{E}[e^{yQ}] - 1}{y} dy \\ &= b^{-1} t(\log t + (1 + \theta) \log \log t) + o(t \log \log t). \end{aligned}$$



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EXAMPLE 3. Let $p \in (0, 1]$, $\mathbb{P}\{Q = b\} = p$ and $\mathbb{P}\{Q \leq 0\} = 1 - p$.
Then $s_\alpha(t)$ admits a representation

$$\begin{aligned}
s_\alpha(t) = & b^{-1} \left(\log(\delta t) + \log \log(\delta t) \right. \\
& + \sum_{n \geq 1} \frac{(-1)^{n+1}}{(\log(\delta t))^n} \sum_{m=1}^n (-1)^{n+m} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} \frac{(\log \log(\delta t))^m}{m!} \Big) \\
& + O((t \log t)^{-1}), \quad t \rightarrow \infty,
\end{aligned}$$

where the series converges for large t , $\delta := (\alpha b p)^{-1}$, and
 $(\begin{bmatrix} i \\ j \end{bmatrix})_{i \geq 1, 1 \leq j \leq i}$ are the unsigned Stirling numbers of the first kind
 $(x(x+1) \cdots (x+i-1) = \sum_{j=1}^i \begin{bmatrix} i \\ j \end{bmatrix} x^j).$

Requires preparatory manipulations and then exploits the **Lagrange inversion formula** as suggested by
de Bruijn (1958) 'Asymptotic methods in analysis'
for obtaining a full asymptotic expansion of the **Lambert function**
(inverse of xe^x).



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EXAMPLE 3 (cont.) Let $p \in (0, 1]$, $\mathbb{P}\{Q = b\} = p$, $\mathbb{P}\{Q \leq 0\} = 1 - p$.

For each $k \in \mathbb{N}$,

$$\alpha \int_0^{s_\alpha(t)} \frac{\mathbb{E}[e^{yQ}] - 1}{y} dy = t \sum_{i=1}^k \frac{(i-1)!}{b^i (s_\alpha(t))^{i-1}} + O\left(\frac{t}{(s_\alpha(t))^k}\right), \quad t \rightarrow \infty.$$

By the theorem, for any $k \in \mathbb{N}$,

$$p_\alpha(t) = \exp\left(-t\left(s_\alpha(t) - \sum_{i=1}^k \frac{(i-1)!}{b^i (s_\alpha(t))^{i-1}} + O\left(\frac{1}{(s_\alpha(t))^k}\right)\right)\right), \quad t \rightarrow \infty.$$



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For instance, for $k = 3$, with $\delta = (\alpha bp)^{-1}$, as $t \rightarrow \infty$,

$$\begin{aligned} p_\alpha(t) = \exp\bigg(& -b^{-1}t\left(\log(\delta t) + \log \log(\delta t) - 1 + \frac{\log \log(\delta t) - 1}{\log(\delta t)} - \right. \\ & \left. + \frac{\log \log(\delta t)(4 - \log \log(\delta t)) - 4}{2(\log(\delta t))^2} \right. \\ & \left. + \frac{\log \log(\delta t)(36 - 15 \log \log(\delta t) + 2(\log \log(\delta t))^2)}{6(\log(\delta t))^3} + O\left(\left(\frac{1}{\log t}\right)^3\right)\right)\bigg). \end{aligned}$$



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EXAMPLE 4. Assume that $Q = b - \eta$, where η is a random variable having the gamma distribution with positive parameters θ and $1/\lambda$. Then $\mathbb{E}[e^{sQ}] = e^{bs}(1 + \lambda s)^{-\theta}$ for $s \geq 0$. In this case, as $t \rightarrow \infty$,

$$\begin{aligned} p_{\alpha}(t) = & \exp \left(-b^{-1}t \left(\log(t/\alpha) + (1 + \theta) \log \log(t/\alpha) - 1 \right. \right. \\ & \left. \left. + (1 + \theta)^2 \frac{\log \log(t/\alpha)}{\log(t/\alpha)} \right. \right. \\ & \left. \left. + \left(\frac{\theta b}{\lambda} + (1 + \theta)^2 \log(1/b) - 1 - \theta \right) \frac{1}{\log(t/\alpha)} + O \left(\left(\frac{\log \log t}{\log t} \right)^2 \right) \right). \end{aligned}$$

and $\mathbb{P}\{Z_{\alpha} > t\}$ admits the same asymptotic expansion.



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Assume that $Q = 1$ a.s. According to the proposition, the distribution of $Z_\alpha - 1$ admits a continuous on $(0, \infty)$ density q_α .

The function q_1^* defined by $q_1^*(t) := e^\gamma q_1(t)$ for $t \geq 0$ is called the **Dickman-de Bruijn function**. Originally appeared in Number Theory and Combinatorics.

Number Theory. The DB function describes the asymptotic distribution of numbers free of large prime factors, so called **smooth or friable** numbers:

$$\lim_{x \rightarrow \infty} \frac{\sum_{n \geq 1} \mathbb{1}_{\{n \leq x^t, P(n) \leq x\}}}{x^t} = q_1^*(t), \quad t \geq 0,$$

where $P(n)$ is the largest prime factor of n .

Combinatorics. Let L_n denote the length of the longest cycle in a uniform random permutation of $\{1, 2, \dots, n\}$. Then

$$\lim_{n \rightarrow \infty} \mathbb{P}\left\{\frac{L_n}{n} \leq x\right\} = q_1^*\left(\frac{1}{x}\right), \quad x > 0.$$



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de Bruijn (1951) used a complex-analytic approach based on the **saddle-point method** to show that

$$q_1(t) \sim \frac{1}{(2\pi t)^{1/2}} \exp\left(-ts_1(t) + \int_0^{s_1(t)} \frac{e^y - 1}{y} dy\right), \quad t \rightarrow \infty,$$

where, for each $t > 1$, $s_1(t)$ is the unique positive solution to $s^{-1}(e^s - 1) = t$.



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where, for each $t > 1$, $s_1(t)$ is the unique positive solution to $s^{-1}(e^s - 1) = t$.

de Bruijn (1951) ($\alpha = 1$) and Vervaat (1972) "Success epochs in Bernoulli trials (with applications in Number Theory)" ($\alpha > 0$) showed that

$$q_\alpha(t) = \exp\left(-t\left(\log t + \log \log t - (1 + \log \alpha)\left(1 + \frac{1}{\log t}\right) + \frac{\log \log t}{\log t} + O\left(\left(\frac{\log \log t}{\log t}\right)^2\right)\right)\right), \quad t \rightarrow \infty.$$



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In a particular case $b = p = 1$, our formula obtained under the assumptions $\mathbb{P}\{Q = b\} = p$ and $\mathbb{P}\{Q \leq 0\} = 1 - p$ recovers the preceding formula and provides a longer expansion.



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where, for each $t > 1$, $s_1(t)$ is the unique positive solution to $s^{-1}(e^s - 1) = t$.

By giving a streamlined (but still complex-analytic) argument, **Tenenbaum** (1995) "Introduction to analytic and probabilistic number theory" obtained

$$q_1(t) = \frac{1}{(2\pi t)^{1/2}} \exp\left(-ts_1(t) + \int_0^{s_1(t)} \frac{e^y - 1}{y} dy\right) \left(1 + O\left(\frac{1}{t}\right)\right)$$

as $t \rightarrow \infty$.



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Goldie & Grübel (1996) gave impetus to further investigations of logarithmic asymptotics of superexponential distribution tails of

- perpetuities - [Hitczenko](#) (2010),
[Hitczenko & Wesolowski](#) (2009), [Kołodziejek](#) (2017, 2018);
- fixed points of inhomogeneous smoothing transforms
 - [Alsmeyer & Dyszewski](#) (2017)



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Assume that M_α has the beta distribution with parameters $\alpha > 0$ and 1, whereas Q is no longer upper bounded.

REGULARLY VARYING TAILS. Assume that

- $\mathbb{E}[\log^+ |Q|] < \infty$;
- $\mathbb{P}\{Q > t\} \sim t^{-\rho} L(t)$ as $t \rightarrow \infty$ for some $\rho > 0$ and some L slowly varying at ∞ ;
- the variables M_α and Q may be dependent.

According to the **Grincevičius-Grey theorem** ([Grincevičius \(1975\)](#), [Grey \(1994\)](#)),

$$\mathbb{P}\{Z_\alpha > t\} \sim \frac{1}{1 - \mathbb{E}[M_\alpha^\rho]} \mathbb{P}\{Q > t\} \sim \frac{\alpha + \rho}{\rho} t^{-\rho} L(t), \quad t \rightarrow \infty.$$



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$$\mathbb{P}\{Z_\alpha > t\} \sim \frac{1}{1 - \mathbb{E}[M_\alpha^\rho]} \mathbb{P}\{Q > t\} \sim \frac{\alpha + \rho}{\rho} t^{-\rho} L(t), \quad t \rightarrow \infty.$$

If M_α and Q are independent, this also follows from

$$\mathbb{P}\{Z_\alpha - Q_1 > t\} \sim \nu_\alpha((t, \infty)) = \alpha \int_t^\infty y^{-1} \mathbb{P}\{Q > y\} dy \sim \frac{\alpha}{\rho} t^{-\rho} L(t)$$

as $t \rightarrow \infty$.



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GAMMA-LIKE TAILS. Assume that

- M_α and Q are independent;
- $\mathbb{E}[\log^+ |Q|] > \infty$;
- $\mathbb{P}\{Q > t\} = Ae^{-ct} + r(t)$ for $t \geq 0$, some $A, c > 0$ and a real-valued function r satisfying several technical conditions.

Then [Buraczewski, Dyszewski, I. & Marynych \(2018\)](#) proved that

$$\mathbb{P}\{Z_\alpha > t\} \sim Bt^{\alpha A}e^{-ct}, \quad t \rightarrow \infty$$

for an explicitly known finite constant B .



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This result is well in line with the fact that if $Q \sim \text{Exp}(c)$, then $Z_\alpha \sim \gamma(\alpha + 1, c)$ and particularly

$$\mathbb{P}\{Z_\alpha > t\} \sim \frac{c^\alpha}{\Gamma(\alpha + 1)} t^\alpha e^{-ct}, \quad t \rightarrow \infty,$$

where Γ is the Euler gamma-function.



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Let $(X(t))_{t \geq 0}$ be a subordinator.

$\hat{Z} := \int_0^\infty e^{-X(u)} du$ is a perpetuity with
 $(M, Q) = (e^{-X(T)}, \int_0^T e^{-X(u)} du)$ for a stopping time T . It is known
that the distribution of $\hat{Z}$ has a density $\hat{p}$. According to a general
theory of perpetuities, $\mathbb{E}[e^{s\hat{Z}}] < \infty$ for some $s > 0$.



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Using a real-analytic argument exploiting a feature of a functional equation satisfied by $t \mapsto (-\log \mathbb{P}\{\hat{Z} > t\})'$, [Haas \(2025\)](#) found a precise asymptotic behavior of $\mathbb{P}\{\hat{Z} > t\}$ and $\hat{p}(t)$ as $t \rightarrow \infty$ under

$$\limsup_{s \rightarrow \infty} s(-\log \mathbb{E}[e^{-sX(1)}])' < 1.$$

With the help of a complex-valued argument based on the saddle-point method, [Minchev & Savov \(2023\)](#) derived similar asymptotics as well as precise asymptotics for the derivatives of $\hat{p}$.



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THANK YOU FOR YOUR ATTENTION!